Evaluating the Real-Time Electrocatalytic Performance of Copper Nanoparticles for CO_2R

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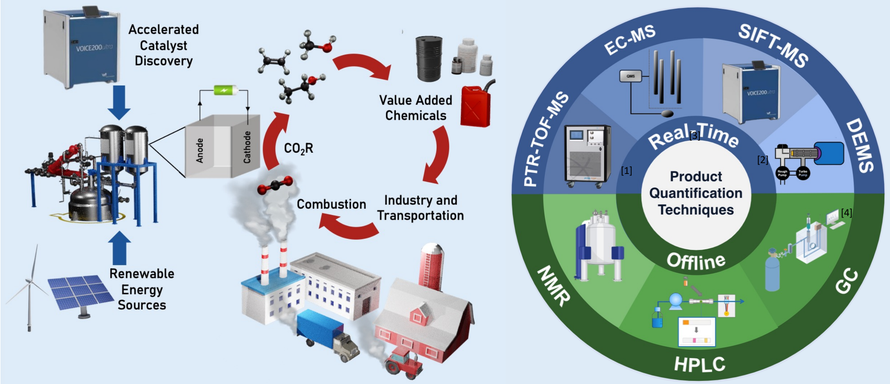
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Background



- CO2R could enable a sustainable energy economy, but requires discovering ideal catalysts
- SIFT-MS is promising for real-time product quantification, accelerating catalyst discovery

Objectives

1.

Determine the real-time catalytic activity of CuNP for CO_2R by obtaining repeatable measurements of FE, PCD, and rate of production for 10 products

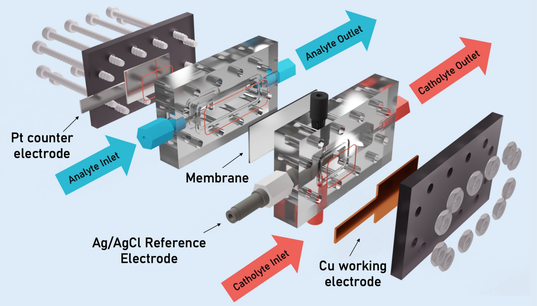
2.

Provide a methodology for evaluating the real-time electrochemical performance of catalysts across a range of electrochemical reactions

3.

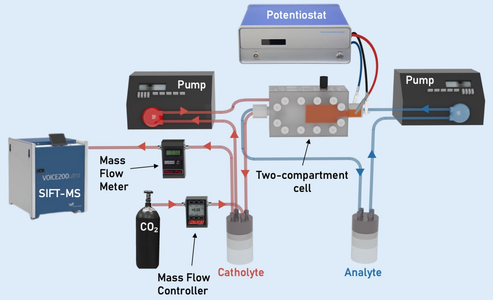
Contribute fundamental knowledge to elucidate the rate-determining steps of C_{2+} products in CO_2R by providing metrics for catalytic performance of CuNP

Methods



Electrochemical Setup

- 2-compartment flow cell
- Reference electrode: Ag/AgCl
- CE: Commercial IrO_2 | WE: CuNP on graphite foil
- Membrane: Selemion anion exchange membrane



Product Quantification:

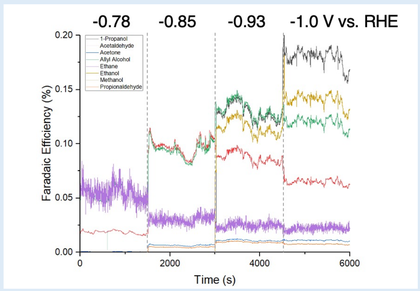
- SIFT-MS

Electrolyte:

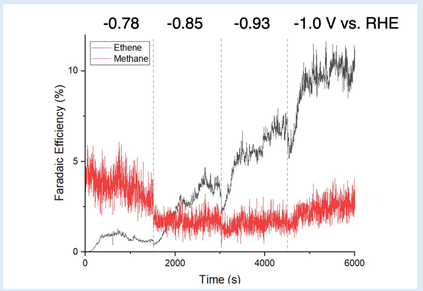
- 0.1 M $KHCO_3$, pH=6.8, purged with CO_2

Results

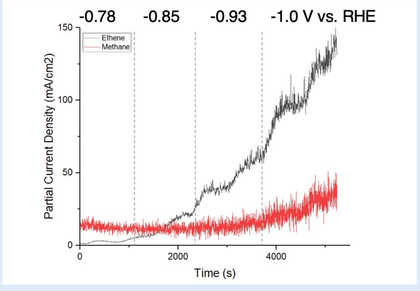
Faradaic Efficiency of Minor Products



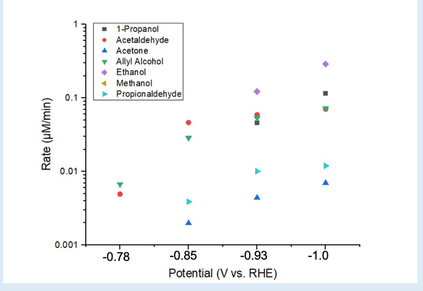
Faradaic Efficiency of Major Products



Partial Current Density



Flow Rate



Conclusions & Outlook

- Able to quantify 10 liquid/gas products in real time to evaluate catalysts
- Developed an experimental methodology applicable to other reactions
- Consistent results between trials
- 1-Propanol, Ethanol, Methanol, and Propionaldehyde below LOD at first potential
- Repeat experiment to obtain third data set and validate results
- Perform acetaldehyde precursor experiment to confirm ethanol production pathway in reaction mechanism
- Use MEA cell for higher product concentrations to determine if any crossover occurs in the analyte

Future Work

4 Publish Our Findings!

3 Acetaldehyde Precursor Experiment

2 Liquid Crossover Analysis

1 3rd Data Set

References

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